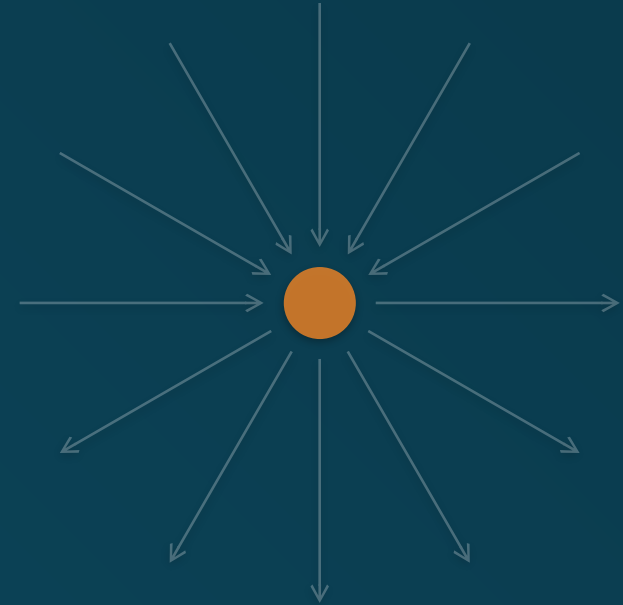


TOPIC 19

Capacitance

电容



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

19.1 Capacitors & capacitance 电容器与电容

- Define capacitance $C = Q/V$ for isolated spheres and parallel-plate capacitors 电容定义
- Derive and use combined capacitance for series and parallel 串并联公式推导与应用

19.2 Energy stored 储能

- Energy from the area under the V-Q graph: $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}Q^2/C$ 储能公式

19.3 Discharging 放电

- Graphs of V, Q, I against time for discharge through a resistor 放电曲线
- Time constant $\tau = RC$ and $x = x_0 e^{-t/RC}$ 时间常数与指数衰减

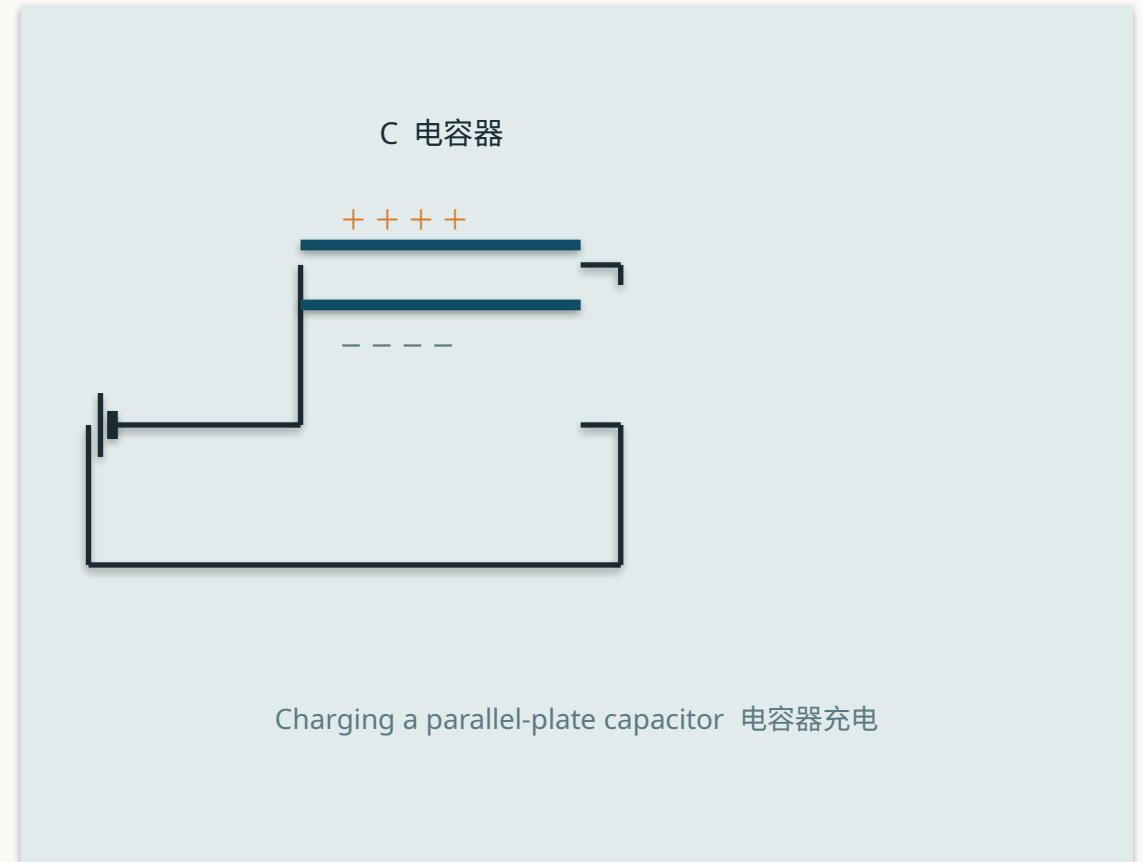
What is Capacitance?

电容的定义

Capacitance C: the charge stored per unit potential difference, . Unit: the **farad (F)**.

单位电势差下储存的电荷量；单位法拉。

- Applies to **parallel-plate capacitors** and to **isolated spherical conductors** 平行板与孤立球导体
- Isolated sphere: （见例题 19A 推导）
- 1 F is huge — practical capacitors are μF , nF, pF 常用微法皮法



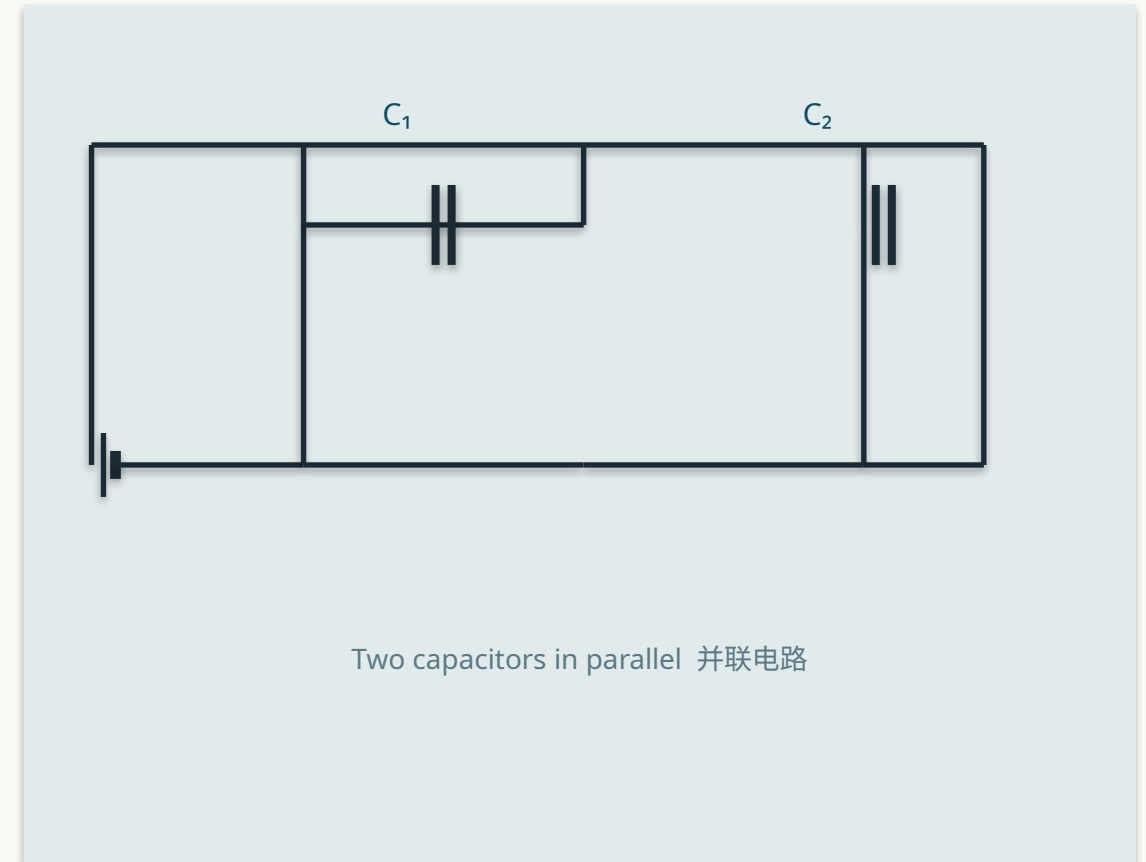
Capacitors in Parallel

电容器的并联

- Same p.d. V across each capacitor 各电容电压相同
- Total charge:

$$C_{total} = C_1 + C_2 \quad \text{parallel 并联等效电容}$$

- Equivalent to one larger capacitor 等效于更大电容
- Compare: same form as resistors in **series** 与电阻串联公式同形



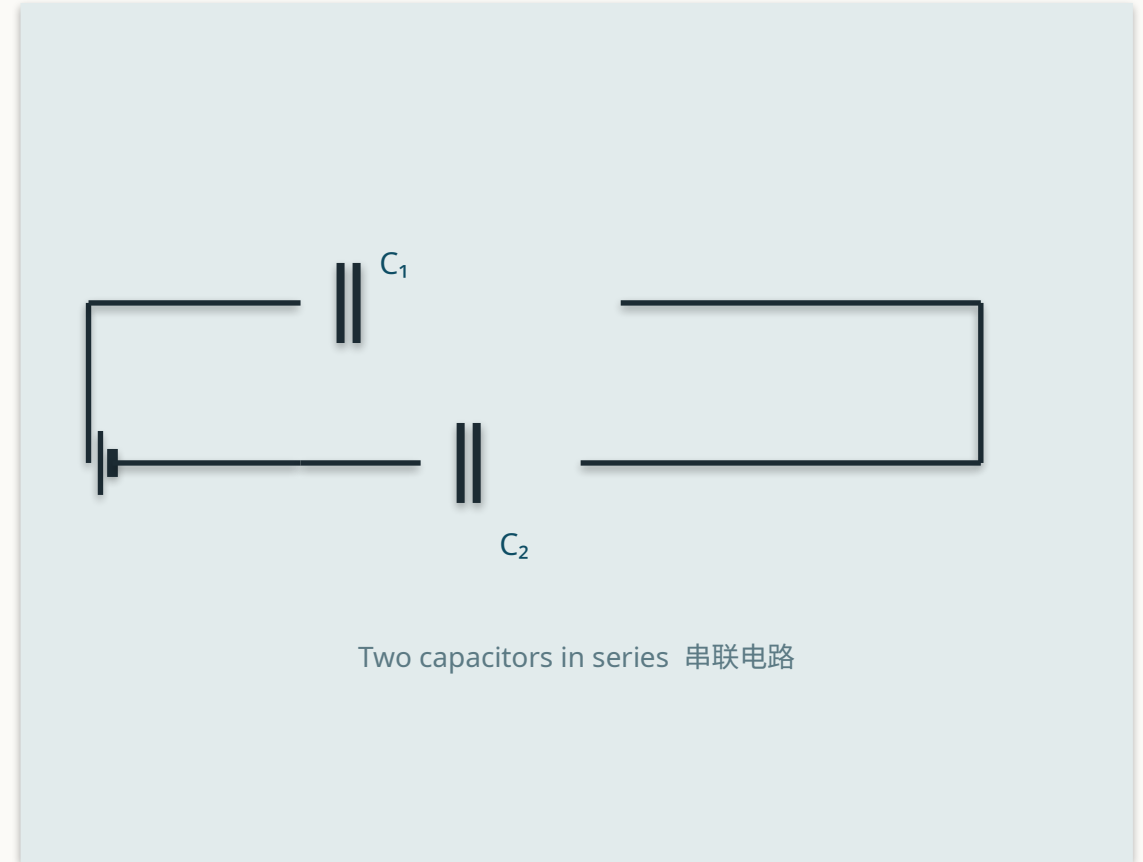
Capacitors in Series

电容器的串联

- Same charge Q on each capacitor 各电容电荷量相同
- Voltages add:

$$\frac{1}{C_{total}} = \frac{1}{C_1} + \frac{1}{C_2} \quad \text{series 串联等效电容}$$

- Result is **smaller** than the smallest capacitor 总电容小于最小者
- Opposite pattern to resistors! 与电阻规律相反
- Two equal C in series give $C/2$ 两个相同电容串联减半



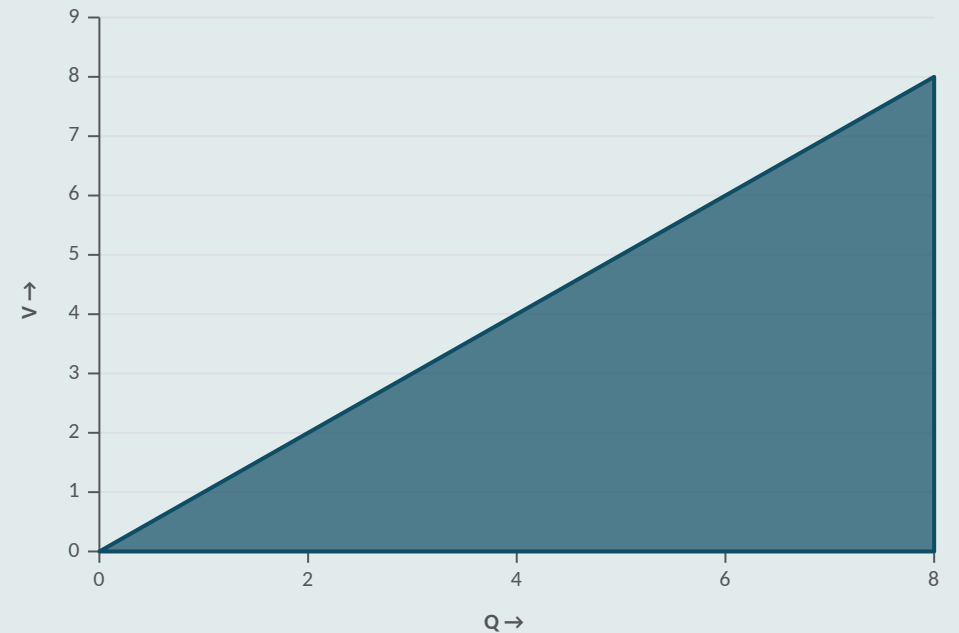
Energy Stored in a Capacitor

电容器的储能

- Charging: the p.d. rises from 0 as charge accumulates — average p.d. is $V/2$ 充电过程电压线性上升
- Energy = area under the V-Q graph 储能 = V-Q 图下方面积

$$W = \frac{1}{2} Q V = \frac{1}{2} C V^2 = \frac{Q^2}{2C} \quad \text{三种等价形式}$$

- Energy is stored in the **electric field** between the plates 能量储存在电场中
- Note the factor $\frac{1}{2}$ — only half the battery's energy is stored 只有一半电源能量被储存

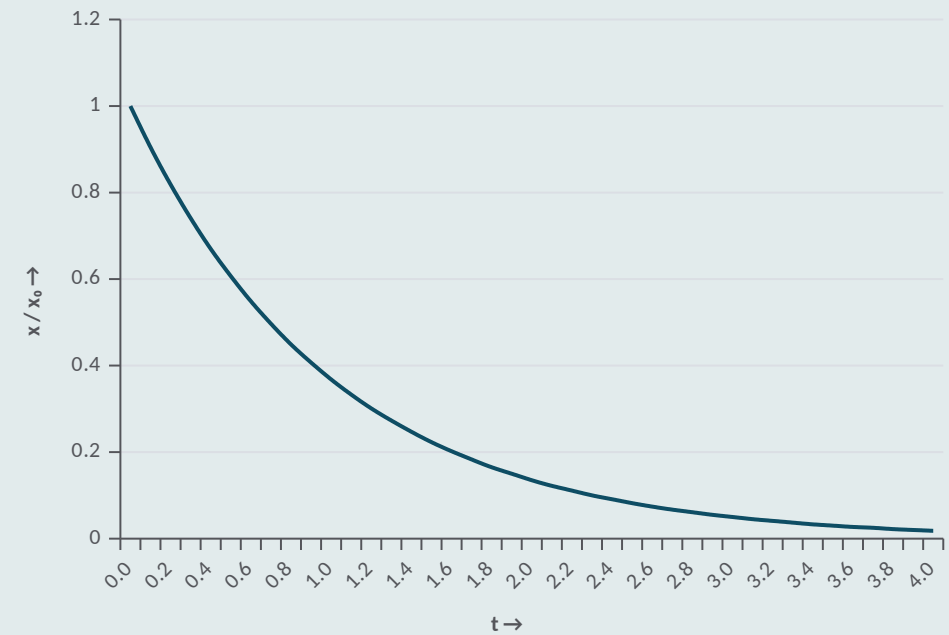


Area under V-Q graph = $\frac{1}{2} QV$ 阴影面积即储能

Discharging a Capacitor

电容器经电阻放电

- Charge, p.d. and current all decay **exponentially** with time
Q、V、I 均指数衰减
- Current is greatest at the start (charge flows fastest) 初始电流最大
- Rate of decay depends on R and C 衰减快慢由 R、C 决定
- The curve approaches zero but never reaches it 渐近为零



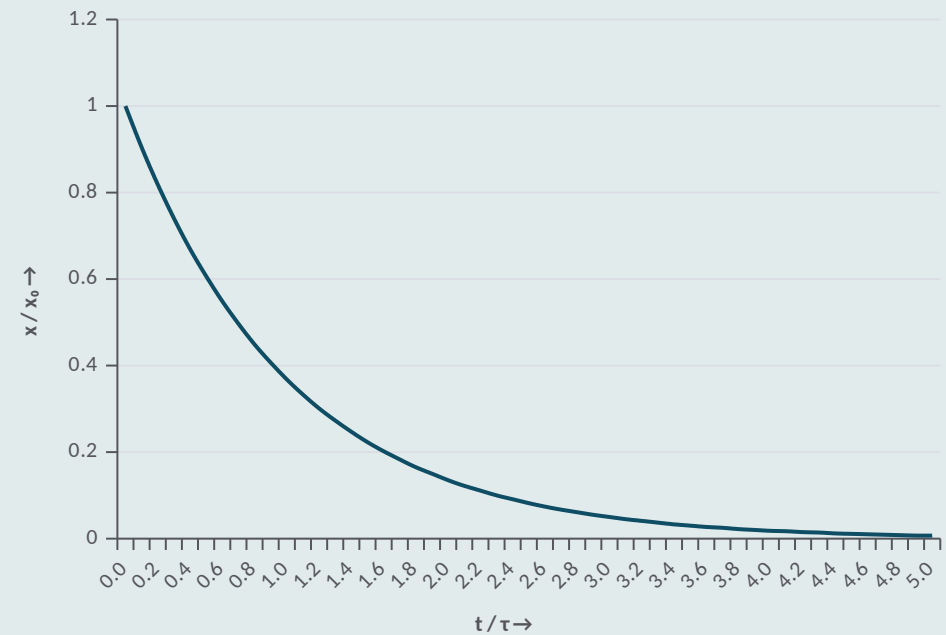
Exponential decay of Q, V and I 指数衰减曲线

The Time Constant

时间常数 $\tau = RC$

$$\tau = RC \quad x = x_0 e^{-t/RC} \quad x \text{ 可为 } Q, V \text{ 或 } I$$

- τ in seconds: R in Ω , C in F 单位秒
- After $t = \tau$: x falls to **37%** ($1/e$) of x_0 一个时间常数后降至 37%
- Half-life 半衰期:
- Large $\tau \Rightarrow$ slow discharge; small $\tau \Rightarrow$ fast discharge τ 大衰减慢



At $t = \tau$, $x = 0.37x_0$ 时间常数的意义

Worked Example 19A

例题解析

WORKED EXAMPLE 19A

1 Show that the capacitance of an isolated spherical conductor of radius r is .

2 Calculate the charge stored on an isolated conductor of capacitance $280\text{ }\mu\text{F}$ at a potential of 25 V .

证明题从定义式出发； μF 要 $\times 10^{-6}$ 。

SOLUTION 解答

Step 1 **1** Charge on a sphere acts as a point charge at its centre: surface potential .

Step 2 By definition .

Step 3 **2**

Answer **2** **(7.0 mC)**

Worked Example 19C

例题解析

WORKED EXAMPLE 19C

1 A $50\ \mu\text{F}$ capacitor in parallel with a $10\ \mu\text{F}$ capacitor is connected to a $12\ \text{V}$ supply. Find (a) total capacitance; (b) p.d. across each; (c) charge on each.

2 A $12\ \mu\text{F}$ capacitor in series with a $6\ \mu\text{F}$ capacitor is connected to $15\ \text{V}$. Find (a) combined capacitance; (b) charge on each; (c) p.d. across each.

串联同 Q 、并联同 V ；最后用 V 总和验证。

SOLUTION 解答

Step 1 **1** Parallel: ; each has $V = 12\ \text{V}$; , .

Step 2 **2** Series: .

Step 3 Same charge on each: .

Step 4 , (check: $5 + 10 = 15\ \text{V}$ ✓)

Answer **1** $60\ \mu\text{F}$; $12\ \text{V}$; $600\ \mu\text{C}$ & $120\ \mu\text{C}$ **2** $4.0\ \mu\text{F}$; $60\ \mu\text{C}$; $5.0\ \text{V}$ & $10\ \text{V}$

Worked Example 19D

例题解析

WORKED EXAMPLE 19D

1 Calculate the energy stored by a $280\ \mu\text{F}$ capacitor charged to $12\ \text{V}$.

2 A camera flash uses a $5000\ \mu\text{F}$ capacitor charged by a $9\ \text{V}$ battery. Calculate the energy transferred when it discharges through the lamp until the p.d. falls to $4.0\ \text{V}$.

部分放电要算能量差，不能直接 $\frac{1}{2}CV^2$ 。

SOLUTION 解答

Step 1 **1.**

Step 2 **2** Energy remaining at $4\ \text{V}$ is **not** transferred — subtract:

Step 3

Answer **1** **2**

Worked Example 19E

例题解析

WORKED EXAMPLE 19E

Calculate the time taken to discharge a $2.5\ \mu\text{F}$ capacitor through a resistance of $0.5\ \text{M}\Omega$ until the p.d. across it has fallen by 50%.

降到 50% 即一个半衰期 $t_{1/2} = 0.693\tau$ 。

SOLUTION 解答

- Step 1** Time constant: .
Step 2 Decay equation: with .
Step 3 Take natural logarithms: .
Step 4

Answer

Class Practice

随堂练习

Q1

The charge on an isolated sphere is 5.4 mC when its potential is 12 V. Calculate its capacitance in microfarads. 球导体电容计算。

Q2

A 470 μF capacitor is charged to 12 V. Calculate (a) the charge stored; (b) the energy stored. 储能与电荷。

Q3

A 100 μF capacitor, initially charged to 9.0 V, discharges through a 47 k Ω resistor. Calculate (a) the time constant; (b) the p.d. after 4.7 s; (c) the p.d. after 9.4 s.

Answers Q1 450 μF Q2 (a) 5.6 mC (b) 0.034 J Q3 (a) 4.7 s (b) 3.3 V (c) 1.2 V

Summary

本章小结

KEY FORMULAS 核心公式

$$C = \frac{Q}{V}$$

definition 电容定义

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

series 串联

$$\tau = RC$$

time constant 时间常数

$$C = C_1 + C_2$$

parallel 并联

$$W = \frac{1}{2} QV = \frac{1}{2} CV^2$$

energy 储能

$$x = x_0 e^{-t/RC}$$

exponential decay 指数衰减

KEY IDEAS 核心思想

- Capacitance links stored charge to p.d. 电容 = 单位电压的储电能力。
- Series/parallel rules are opposite to those for resistors 与电阻规律相反。
- Discharge is exponential; after τ the value falls to 37% 放电指数衰减。

Exam Tips

备考提示

- Series/parallel capacitor rules are the **reverse** of resistor rules — check twice. 串并联规则与电阻相反。
- $\mu\text{F} = 10^{-6} \text{ F}$; $\text{k}\Omega = 10^3 \Omega$ — τ comes out in seconds automatically if SI units are used. 单位换算。
- For partial discharge, compute the energy **difference**, not $\frac{1}{2}CV^2$. 部分放电算差值。
- Use $\ln$ when solving for t ; remember $50\% \Rightarrow t = 0.693\tau$. 指数方程取对数。